Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

If the mean value theorem is $f(b) - f(a) = (b-a)f'(c)$. Then, for the function $x^2 - 2x + 3$ in $\left[1, \frac{3}{2}\right]$ the value of c is

(A) $6/5$

(B) $5/4$

(C) $4/3$

(D) $7/6$

Q.2

Which of the following is correct for $f(x) = \begin{cases} (x-e)2^{-2^{\left(\frac{1}{(e-x)}\right)}}, & x \neq e \text{ at } x = e \\ 0, & x = e \end{cases}$

(A) $\mathrm{f(x)}$ is discontinuous at $\mathrm{x = e}$

(B) $f(x)$ is differentiable at $x = e$

(C) $\mathrm{f(x)}$ is non-differentiable at $\mathrm{x = e}$

(D) None of these

Q.3

If $y = \tan^{-1}\left(\frac{2^x}{1+2^{2x+1}}\right)$, then $\frac{dy}{dx}$ at $x = 0$ is

(A) $\frac{3}{5}\log 2$

(B) $\frac{2}{5}\log 2$

(C) $-\frac{3}{2}\log 2$

(D) $\log 2\left(\frac{-1}{10}\right)$

Q.4

If $f(x) = \log_x(\ln x)$, then at $x = e$, $f'(x)$ equals-

(A) 0

(B) 1

(C) e

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(D) $1/e$

Q.5

$s = \sqrt{t^2 + 1}$, then $\frac{d^2 s}{dt^2}$ is equal to

(A) $\frac{1}{s}$

(B) $\frac{1}{s^2}$

(C) $\frac{1}{s^3}$

(D) $\frac{1}{s^4}$

Q.6

What is the derivative of $\tan^{-1}\left(\frac{\sqrt{1+x^2}-1}{x}\right)$ with respect to $\tan^{-1} x$?

(A) 0

(B) $\frac{1}{2}$

(C) 1

(D) x

Q.7

Let $g(x) = f(x)\sin x$, where $f(x)$ is a twice differentiable function on $(-\infty, \infty)$ such that $f'(-\pi) = 1$. The value of $g''(-\pi)$ equals

(A) 1

(B) 2

(C) -2

(D) 0

Q.8

Let $f(x) = [n + p\sin x], x \in (0, \pi), n \in Z$ and P is a prime number, where $[\cdot]$ denotes the greatest integer function. Then number of points where $f(x)$ is not differentiable is

(A) $2p + 1$

(B) $2p - 1$

(C) $2p$

(D) $2p - 2$

Q.9

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A function $f : \mathbf{R} \to \mathbf{R}$ is differentiable and satisfies the condition $f\left(\frac{1}{n}\right) = 0$ for all integers $n \geq 1$. Then

(A) $f(x) = 0$ for all $x \in (0, 1]$

(B) $f(0) = 0 = f'(0)$

(C) $f(0) = 0$ but $f'(0)$ need not be equal to 0 .

(D) $|f(x)| \leq 1$ for all $x \in [0, 1]$

Q.10

Let $f(x)$ and $g(x)$ be differentiable functions for $0 \leq x \leq 1$ such that $f(0) = 2, g(0) = 0, f(1) = 6$ Let there exist a real number c in $[0, 1]$ such that $f'(\mathrm{c}) = 2\ \mathrm{g}'(\mathrm{c})$, then the value of $\mathrm{g}(1)$ must be

(A) 1

(B) 2

(C) -2

(D) -1

Q.11

If $f(x)$ is a differentiable function and the curve $y = f(x)$ passes through $(1, 1)$ such that $2f(x + y) + f(x - y) + 3y^2 = 3f(x) + 2xy$, then $\lim_{x \to 1} \frac{f(x) - 1}{x - 1}$ is equal to

(A) 3

(B) 0

(C) 2

(D) 1

Q.12

The function $'f'$ is differentiable $\forall x \geq 0$ and $f'(x) \sin x + f(x) \cos x \leq 0$. Then range of values of x, if $f(x) > 0 \quad \forall x \geq 0$ is,

(A) $[0, \infty)$

(B) $[2\mathrm{n}\pi + \pi, 2\mathrm{n}\pi + 2\pi], \mathrm{n} \in \mathrm{Z}_+$

(C) $[2n\pi, 2n\pi + \pi], n \in z_+$

(D) none of these

Q.13

The function $f(x) = \left(x^2 - 1\right) \left|x^2 - 3x + 2\right| + \cos(|x|)$ is not differentiable at

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(A) -1

(B) 0

(C) 1

(D) 2

Q.14

The number of points of non differentiability of $y = |\log|x||$ are

(A) 3

(B) 1

(C) 2

(D) none of these

Q.15

Let $f : \mathrm{R} \to \mathrm{R}$ be a function defined by $f(\mathrm{x}) = \max\left(x, x^3\right)$. The set of all points where $f(x)$ is not differentiable is

(A) $\{-1, 1\}$

(B) $\{-1, 0\}$

(C) $\{0, 1\}$

(D) $\{-1, 0, 1\}$

Q.16

If $f(x) = \begin{cases} |x-4|, \text{ for } x \geq 1 \\ \left(\frac{x^3}{2} - x^2 + 3x + \frac{1}{2}\right) \text{ for } x < 1, \text{ then} \end{cases}$

(A) $f(x)$ is continuous at $x = 1$ and at $x = 4$

(B) $f(x)$ is differentiable at $x = 1$

(C) $f(x)$ is continuous and differentiable at $x = 1$

(D) $f(x)$ is only continuous at $x = 1$

Q.17

Let $f(x)$ be a function differentiable at $x = c$. Then $\lim_{x \to c} f(x)$ equals

(A) $f(\mathrm{c})$

(B) $f'(\mathrm{c})$

(C) $f''(\mathrm{c})$

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(D) $1/f(c)$

Q.18

If $f(x) = x(\sqrt{x} - \sqrt{x+1})$, then

(A) f is continuous but not differentiable at $x = 0$

(B) f is differentiable at $x = 0$.

(C) f is differentiable but not continuous at $x = 0$

(D) f is not differentiable at $x = 0$

Q.19

Let f and g be differentiable functions satisfying $g'(a) = 2, g(a) = b$ and $fog = I$ (Identity function). Then $f'(b)$ is equal to

(A) 0

(B) $1/2$

(C) 1

(D) 2

Q.20

Let $f(x) = \frac{\sin 4\pi[x]}{1+[x]^2}$, where $[x]$ is the greatest integer less than or equal to x, then

(A) $f(x)$ is not differentiable at some points

(B) $f'(x)$ exists but is different from zero

(C) $f'(x) = 0$ but f is not a constant function

(D) $f'(x) = 0$ for all x

Answers & Solutions

Q.1 Answer:

$5/4$

Solution:

Let $f(x) = x^2 - 2x + 3$

Since, $f'(c) = \frac{f\left(\frac{3}{2}\right) - f(1)}{\frac{3}{2} - 1}$ **(given)**

$\Rightarrow 2c - 2 = \frac{\frac{9}{4} - \frac{6}{2} + 3 - (1-2+3)}{\frac{3}{2}-1} \Rightarrow c = \frac{5}{4}$

Q.2 Answer:

$f(x)$ is non-differentiable at $x = e$

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Solution:

$$RHD = \lim_{h\to 0}\frac{f(e+h)-f(e)}{h}$$
$$= \lim_{h\to 0}\frac{f(e+h)-0}{h}$$
$$= \lim_{h\to 0}\frac{(e+h-e)2^{-2^{\left(\frac{1}{e-e-h}\right)}}}{h}$$
$$= \lim_{h\to 0}\frac{h\cdot 2^{-2^{\left(-\frac{1}{h}\right)}}}{h}$$
$$= 2^{-2^{\left(\frac{1}{-0}\right)}}$$
$$= 2^{-2^{-\infty}} = 2^{-0} = 2^0 = 1$$

$$LHD = \lim_{h\to 0}\frac{f(e-h)-f(e)}{-h} = \lim_{h\to 0}\frac{f(e-h)}{-h}$$
$$= \lim_{h\to 0}\frac{(e-h-e)2^{-2^{\left(\frac{1}{e-e+h}\right)}}}{}$$
$$= \lim_{h\to 0}\frac{-\ h\cdot 2^{-2^{\left(\frac{1}{h}\right)}}}{-h}$$
$$= 2^{-2^{\infty}} = 0$$

$RHD \neq LHD$

$f(x)$ **is not differentiable at** $x = e$

Q.3 Answer:

$\log 2\left(\frac{-1}{10}\right)$

Solution:

Given expression can be written as $y = \tan^{-1}\left[\frac{2^x(2-1)}{1+2^x\cdot 2^{x+1}}\right] = \tan^{-1}\left[\frac{2^{x+1}-2^x}{1+2^x\cdot 2^{x+1}}\right]$

$= \tan^{-1}\left(2^{x+1}\right) - \tan^{-1}\left(2^x\right)$

$\Rightarrow \frac{dy}{dx} = \frac{2^{x+1}\log 2}{1+2^{2(x+1)}} - \frac{2^x\log 2}{1+2^{2x}}$

$\therefore \left(\frac{dy}{dx}\right)_{x=0} = (\log 2)\left(\frac{2}{5}-\frac{1}{2}\right) = \log 2\left(-\frac{1}{10}\right)$

Q.4 Answer:

$1/e$

Solution:

$\because \ln x = \log_e x$, **so**

$f(x) = \log_x(\log_e x) = \frac{\log(\log x)}{\log x}$

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$\Rightarrow f'(x) = \frac{\log x\left(\frac{1}{x\log x}\right) - \log(\log x)\cdot\frac{1}{x}}{(\log x)^2}$

$\therefore f'(e) = \frac{1/e - 0}{(1)^2} = \frac{1}{e}$

Q.5 Answer:

$\frac{1}{s^3}$

Solution:

$$s = \sqrt{t^2+1}$$

$$\Rightarrow \frac{ds}{dt} = \frac{t}{\sqrt{t^2+1}} \quad \Rightarrow \frac{d^2 s}{dt^2} = \frac{1}{\sqrt{(t^2+1)^3}}$$

$$\Rightarrow \frac{d^2 s}{dt^2} = \frac{1}{s^3}$$

Q.6 Answer:

$\frac{1}{2}$

Solution:

Let $y = \tan^{-1}\left[\frac{\sqrt{1+x^2}-1}{x}\right]$ **and** $u = \tan^{-1} x$

Put $x = \tan\theta \Rightarrow \theta = \tan^{-1} x$

Then, $y = \tan^{-1}\left[\frac{\sqrt{1+\tan^2\theta}-1}{\tan\theta}\right]$

$= \tan^{-1}\left[\frac{\sqrt{\sec^2\theta}-1}{\tan\theta}\right]$

$= \tan^{-1}\left[\frac{\sec\theta-1}{\tan\theta}\right] = \tan^{-1}\left[\frac{\frac{1}{\cos\theta}-1}{\frac{\sin\theta}{\cos\theta}}\right]$

$= \tan^{-1}\left[\frac{1-\cos\theta}{\sin\theta}\right] = \tan^{-1}\left[\frac{2\sin^2\frac{\theta}{2}}{2\sin\frac{\theta}{2},\cos\frac{\theta}{2}}\right]$

$\left(\begin{array}{c} \because 1-\cos\theta = 2\sin^2\frac{\theta}{2} \text{ and} \\ \sin x = 2\sin\frac{x}{2}\cdot\cos\frac{x}{2} \end{array}\right)$

$= \tan^{-1}\left[\tan\frac{\theta}{2}\right]$

$\Rightarrow \quad y = \frac{\theta}{2} \Rightarrow y = \frac{\tan^{-1} x}{2} \quad \left[\because \theta = \tan^{-1} x\right]$

$\Rightarrow \quad y = \frac{u}{2}; \quad \frac{dy}{du} = \frac{1}{2}$

Q.7 Answer:

-2

Solution:

We have $g(x) = f(x)\sin x \ldots (1)$

On differentiating equation (1) w.r.t. x**,**

we get

$g'(x) = f(x)\cos x + f'(x)\sin x \ldots..(2)$

Again differentiating equation (2) w.r.t. x**, we get**

$g''(x) = f(x)(-\sin x) + f'(x)\cos x + f'(x)\cos x + f''(x)\sin x \ldots\ldots(3)$

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$\Rightarrow g''(-\pi) = 2f'(-\pi)\cos(-\pi) = 2 \times 1 \times -1 = -2$

Hence $g''(-\pi) = -2$

Q.8 Answer:

$2\mathrm{p} - 1$

Solution:

$f(x)$ **is not differentiable at those points where** $\mathrm{n} + \mathrm{p}\sin x$ **is integer. As** p **is a prime number.**

$\Rightarrow \mathrm{n} + \mathrm{p}\sin x$ **is an integer if** $\sin x = 1, -1, \mathrm{r}/\mathrm{p}$

i.e., $x = \frac{\pi}{2}, -\frac{\pi}{2}, \sin^{-1}\frac{r}{p}, \pi - \sin^{-1}\frac{r}{p}$

Where $0 \leq r \leq p - 1$

But $x \neq -\frac{\pi}{2}, 0$

Function is not differentiable at $x = \frac{\pi}{2}, \sin^{-1}\frac{r}{p}, \pi - \sin^{-1}\frac{r}{p}$ **where** $0 < r \leq p - 1$

So the number of points is $= 1 + 2(p - 1) = 2p - 1$

Q.9 Answer:

$f(0) = 0 = f'(0)$

Solution:

$x_n = \frac{1}{n}(n \in N) \to 0$ **asn** $\to \infty$**. By the definition of continuity,** $f(x_n) \to f(0)$ **as** $n \to \infty$

But $f(x_n) = 0$ **for all** $x \in N$**. So** $f(0) = 0$**.**

Also $\underset{x\to 0}{Lt}\frac{f(x)-f(0)}{x-0} = \underset{x\to 0}{Lt}\frac{f(x_n)-f(0)}{x_n-0}(\because f$ **is derivable at** $x = 0)$

Q.10 Answer:

2

Solution:

$\frac{f(1)-f(0)}{1-0} = f'(c)$ **and** $\frac{g(1)-g(0)}{1-0} = g'(c)$

$\Rightarrow \frac{f(1)-f(0)}{g(1)-g(0)} = \frac{f'(c)}{g'(c)}$

$\Rightarrow \frac{6-2}{g(1)-0} = \frac{2g'(c)}{g'(c)} = \frac{2}{1} \Rightarrow 4 = 2g(1) \Rightarrow g(1) = 2$

Q.11 Answer:

2

Solution:

$\lim_{x\to 1}\frac{f(x)-1}{x-1}\left(\frac{0}{0}\right) = f'(1)$

Now put $x = 1$ **and** $y = h$ **in the given relation**

$2\frac{f(1+h)-f(1)}{h} - \frac{f(1-h)-f(1)}{-h}3h = 2$

talking limit $\mathrm{h} \to 0 \quad 2f'(1) - f'(1) = 2$

$f'(1) = 2$

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Q.12 Answer:

$[2n\pi + \pi, 2n\pi + 2\pi], n \in Z_+$

Solution:

$f'(x)\sin x + f(x)\cos x \leq 0 \forall x \geq 0$
$\Rightarrow \frac{d}{dx}[f(x)\sin x] \leq 0$
$g(x) = f(x)\sin x \forall x \geq 0$
$\Rightarrow x \geq 0 \quad g(x) \leq g(0)$
$\Rightarrow f(x)\sin x \leq 0 \quad \Rightarrow \sin x \leq 0$
$x \in [2n\pi + \pi, 2n\pi + 2\pi] n \in z_+$

Q.13 Answer:

2

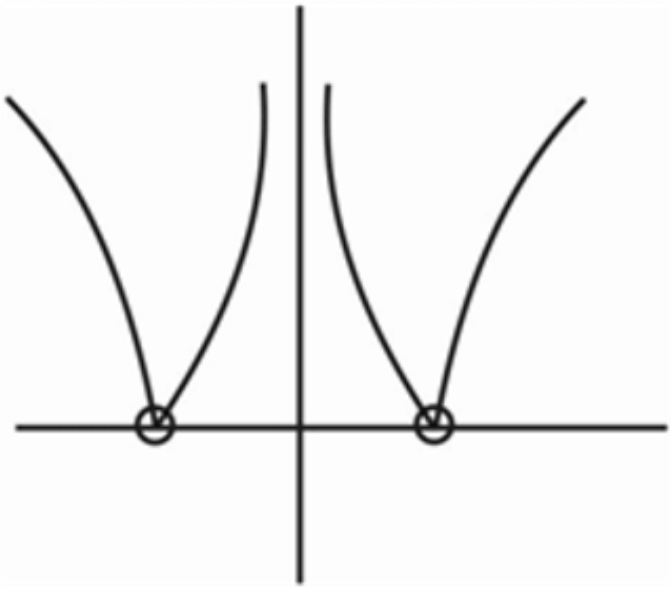
Solution:

$x^2 - 3x + 2 = (x-1)\cdot(x-2) > 0$ **for** $x < 1$ **or** $x > 2$ **and is** ≤ 0 **when** $1 \leq x \leq 2$.
Also, $\cos|x| = \cos x (\because \cos(-x) = \cos x)$
$\therefore f(x) = \begin{cases} -(x^2-1)(x^2-3x+2) + \cos x, 1 \leq x \leq 2 \\ (x^2-1)(x^2-3x+2) + \cos x, x > 2 \end{cases}$
Clearly, $f(x)$ **is not differentiable at** $x = 2$ **as** $L' \neq R'$

Q.14 Answer:

2

Solution:



Q.15 Answer:

$\{-1, 0, 1\}$

Solution:

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If $x < -1$ then $x > x^3$. So, $f(x) = x$,
If $x = -1$ then $x = x^3$. So, $f(x) = x$,
If $-1 < x < 0$ then $x < x^3$. So, $f(x) = x^3$,
If $x = 0$ then $x = x^3$. So, $f(x) = x^3$,
If $0 < x < 1$ then $x > x^3$. So, $f(x) = x$,
If $x = 1$ then $x = x^3$. So, $f(x) = x$,
If $x > 1$, then $x < x^3$. So, $f(x) = x^3$

Thus $f(x) = x, x \leq -1, f(x) = \begin{cases} x & , x \leq -1 \\ x^3 & , -1 < x \leq 0 \\ x & , 0 < x \leq 1 \\ x^3 & , \quad x > 1 \end{cases}$ **Clearly,** $f(x)$ **is not differentiable**

at $x = -1$,
0,1

Q.16 Answer:

$f(x)$ is continuous at $x = 1$ and at $x = 4$

Solution:

We have,

$$f(x) = \begin{cases} x - 4, x \geq 4 \\ -(x-4), 1 \leq x < 4 \\ \left(\frac{x^3}{2} - x^2 + 3x + \frac{1}{2}\right), x < 1 \end{cases}$$

Clearly, $\mathrm{f(x)}$ **is continuous for all**
x **but it is not differentiable at** $x = 1$ **and** $x = 4$.

Q.17 Answer:

$f(\mathrm{c})$

Solution:

Since $f(x)$ **is differentiable at** $x = c$**, therefore it is continuous at** $x = c$.
Hence, $\lim_{x \to c} f(x) = f(c)$

Q.18 Answer:

f is differentiable at $x = 0$.

Solution:

$$f(x) = x(\sqrt{x} - \sqrt{x+1})$$

We have, $\Rightarrow \quad f'(x) = (\sqrt{x} - \sqrt{x+1}) + \dfrac{x\sqrt{x+1} - \sqrt{x}}{2\sqrt{x} \times \sqrt{x+1}}$

for $x \neq 0$

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Also $f'(0^+) = -1$

So, $f(x)$ **is differentiable and hence, continuous at** $x = 0$**.**

Q.19 Answer:

$1/2$

Solution:

$$f \circ g = I$$

$$\Rightarrow f \circ g(x) = x \text{ for all } x$$

$$\Rightarrow f'(g(x))g'(x) = 1 \text{ for all } x$$

Given that, $\Rightarrow \quad f'(g(a)) = \dfrac{1}{g'(a)}$

$$= \frac{1}{2}$$

$$\Rightarrow f'(b) = \frac{1}{2}. \quad [\because g(a) = b]$$

Q.20 Answer:

$f'(x) = 0$ for all x

Solution:

We have,

$$f(x) = \frac{\sin 4\pi[x]}{1 + [x]^2} = 0$$

for all $x[\because 4\pi[x]$ **is a multiple of** $\pi] \Rightarrow f'(x) = 0$ **for all** x**.**

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